# Programme: 6

## Application to Chemical Bonding

7 Marks

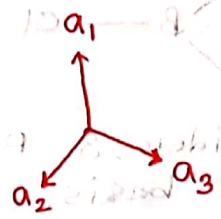
6.1 Group theory summerised in three rules:

(i) Use an appropriate basis to generate a reducible representation of the point group.

(ii) Reduce this representation to its constituent irreducible representations.

(iii) Interpret the results.

6.4:




$D_{3h}$

character = the number of vectors unsifted by the operation.

E, character = 3 ; All vectors remain unfhifted.

$C_3$, " = 0 ; " " are shifeted.

$C_2$ " = 1 ; 1 unchanged, 2 changed.

$\sigma_h$ " = 3 ; All unchanged.

$\sigma_v$ " = 1 ; 1 unchanged.

~~Alt~~ The complete set of characters is thus,

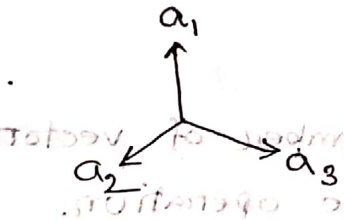
| $D_3$ | E | $2C_3$ | $3C_2$ | $\sigma_h$ | $2S_3$ | $3\sigma_v$ | |
|---|---|---|---|---|---|---|---|
| $\Gamma_1$ | 3 | 0 | 1 | 3 | 0 | 1 | → Reducible representation. |

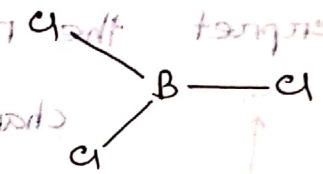
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## Character table

| $D_{3h}$ | E | $2C_3$ | $3C_2$ | $\sigma_h$ | $2S_3$ | $3\sigma_v$ | | |
|---|---|---|---|---|---|---|---|---|
| $A_1'$ | 1 | 1 | 1 | 1 | 1 | 1 | | $x^2+y^2, z^2$ |
| $A_2'$ | 1 | 1 | −1 | 1 | 1 | −1 | $R_z$ | |
| $E'$ | 2 | −1 | 0 | 2 | −1 | 0 | $(x,y)$ | $(x^2-y^2, xy)$ |
| $A_1''$ | 1 | 1 | 1 | −1 | −1 | −1 | | |
| $A_2''$ | 1 | 1 | −1 | −1 | −1 | 1 | $z$ | |
| $E''$ | 2 | −1 | 0 | −2 | 1 | 0 | $(R_x, R_y)$ | $(xz, yz)$ |




consider 3 B-Cl bond as a basis.

| $D_{3h}$ | E | $2C_3$ | $3C_2$ | $\sigma_h$ | $2S_3$ | $3\sigma_v$ |
|---|---|---|---|---|---|---|
| $\Gamma_1$ | 3 | 0 | 1 | 3 | 0 | 1 |

→ [symmetry operation করলে যে অক্ষুন্নত point unshifted থাকে]

No. of $A_1' = \frac{1}{12}[(3\times1\times1)+0+(1\times1\times3)+(3\times1\times1)+0+(1\times1\times3)] = 1$

No. of $A_2' = \frac{1}{12}[(3\times1\times1)+0+(1\times-1\times3)+(3\times1\times1)+0+(1\times-1\times3)] = 0$

No. of $E' = \frac{1}{12}[(3\times2\times1)+0+(1\times0\times3)+(3\times2\times1)+0+0] = 1$

No. of $A_1'' = \frac{1}{12}[(3\times1\times1)+0+(1\times1\times3)+(3\times-1\times1)+0+(1\times-1\times3)] = 0$

No. of $A_2'' = \frac{1}{12}[(3\times1\times1)+0+(1\times-1\times3)+(3\times-1\times1)+0+(1\times1\times3)] = 0$

No. of $E'' = \frac{1}{12}[(3\times2\times1)+0+(1\times0\times3)+(3\times-2\times1)+0+(1\times0\times3)] = 0$

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$\therefore \Gamma_1 = A_1' + E'$ [$A_1'$ = single degenerate, $E'$ = Double "] total : 3 for 3 ✓

→ From the character table we see that, s-orbital belongs to the symmetry species $A_1'$ [bcz all operation has only 1]

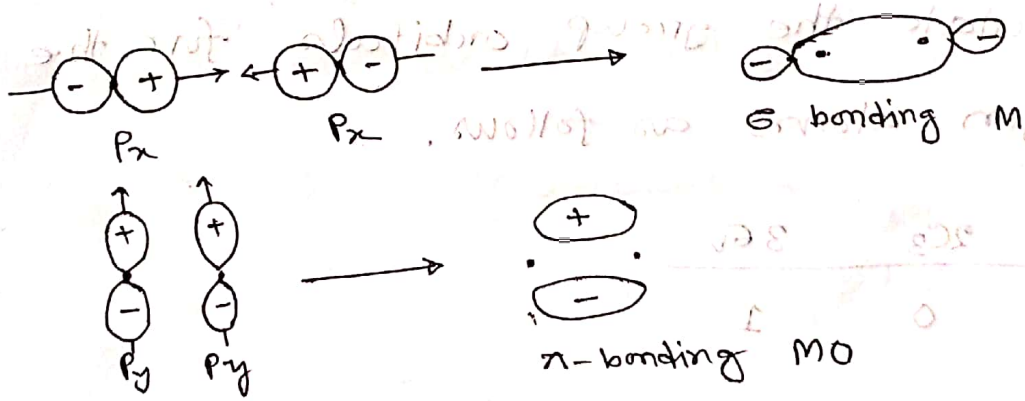
→ $P_x$ & $P_y$ orbitals belongs to the symmetry speciers $E'$

→ $d_{x^2-y^2}$ and $d_{xy}$ orbital belong to the symmetry speciers $E'$. But in case of B, $d_{x^2-y^2}$ & $d_{xy}$ orbitals are energitically inaccessable.

[N.B: $P_z$ here unhybridyzed. So the most likely, set of hybrid orbitals for B in $BCl_3$ is $sp^2$.

**01: Question for final :**

* We know for the p-orbitals wave function.




Book : 106 - 109 02/12/2021 ; 05/12/2021

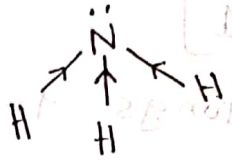
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* $NH_3$

i) The structure of $NH_3$ is Trigonal pyramidal.

ii) Structure:



iii) Ammonia belongs to the point group $C_{3v}$

For N

iv) Among the 4 valence shell orbitals of Nitrogen

2s orbital belongs to $A_1$ representation.

$2p_x$ and $2p_y$ " " " E " (Degenerate).

$2p_z$ " " " $A_1$ "

For 3H

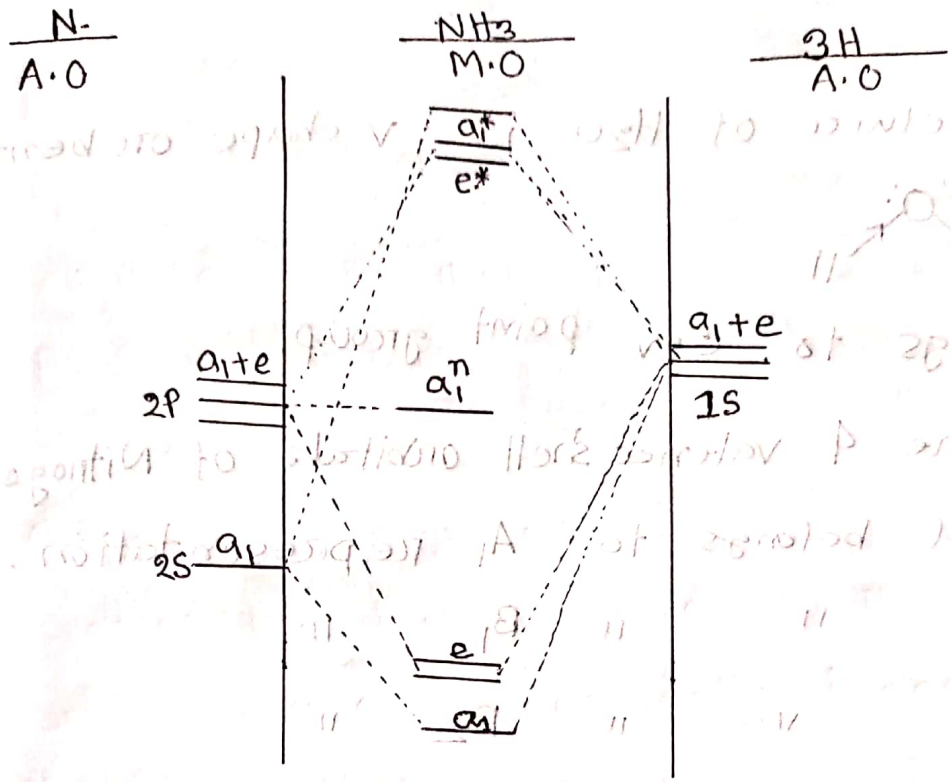
By taking the 3 N-H bond on the basis, we can calculate the group orbitals for the 3 hydrogen atoms as follows.

| $C_{3v}$ | E | $2C_3$ | $3\sigma_v$ |
|---|---|---|---|
| $\Gamma$ | 3 | 0 | 1 |

No. of $A_1 = 1$, $A_2 = 0$, $E = 1$.

So, $\Gamma = A_1 + E$ [Here total '3' orbitals present].

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In $NH_3$, N has 5 electrons in its valence shell and 3-hydrogen atom have 3 electrons.

NOw, we can put these 5+3=8 electrons into the molecular orbitals as follows.

$a_1^*$, $e^*$ } antibonding

$a_1^n$ } non-bonding

e, $a_1$ } bonding

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## $H_2O$

The structure of $H_2O$ is V-shape or bent.



$H_2O$ belongs to $C_{2v}$ point group.

Among the 4 valence shell orbitals of Nitrogen

2s orbital belongs to $A_1$ representation.

$2P_x$ " " " $B_1$ "

$2P_y$ " " " $B_2$ "

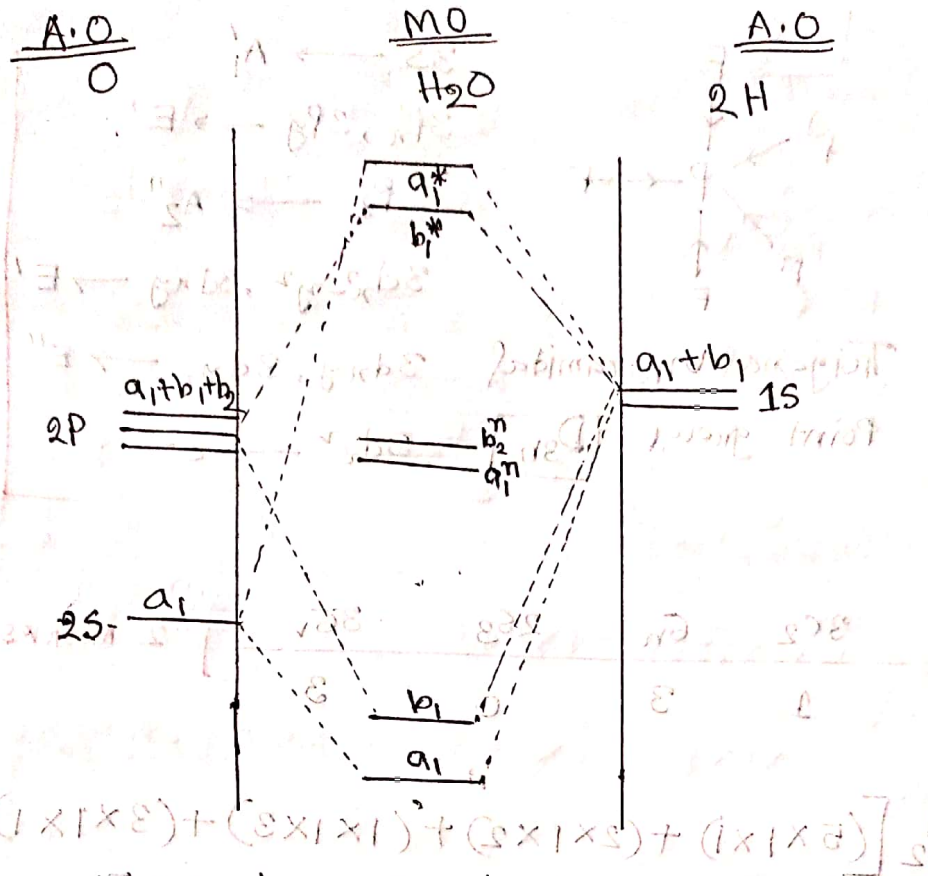
$2P_z$ " " " $A_1$ "

By taking 2 O-H bond on the basis, we can calculate the group orbitals for the 2 hydrogen atoms as follows.

| $C_{2v}$ | E | $C_2$ | $\sigma_v(xy)$ | $\sigma_v'(yz)$ |
|---|---|---|---|---|
| $\Gamma$ | 2 | 0 | 2 | 0 |

No. of $A_1 = 1$ ; $A_2 = 0$ ; $B_1 = 1$ , $B_2 = 0$

So, $\Gamma = A_1 + B_1$ .




In $H_2O$, O has 6 electron in its valence shell, and 2 Hydrogen atom have 2 electrons.

Now, we can put these $6+2=8$ electrons into the molecular orbitals as follows.

$a_1^*$
$b^*$ } Anti-bonding

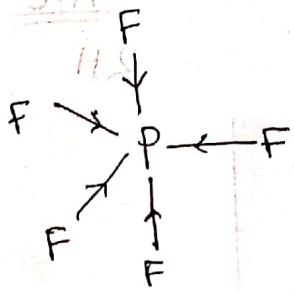
$b_2^n$
$a_1^n$ } non-bonding

$b_1$
$a_1$ } bonding.

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$PF_5$




Trigonal bipyramidal

Point group $D_{3h}$

$3s \longrightarrow A_1'$

$3p_x, 3p_y \longrightarrow E'$

$3p_z \longrightarrow A_2''$

$3d_{x^2-y^2}, 3d_{xy} \longrightarrow E'$

$3d_{xy}, 3d_{yz} \longrightarrow E''$

$3d_{z^2} \longrightarrow z^2$

1 Mark

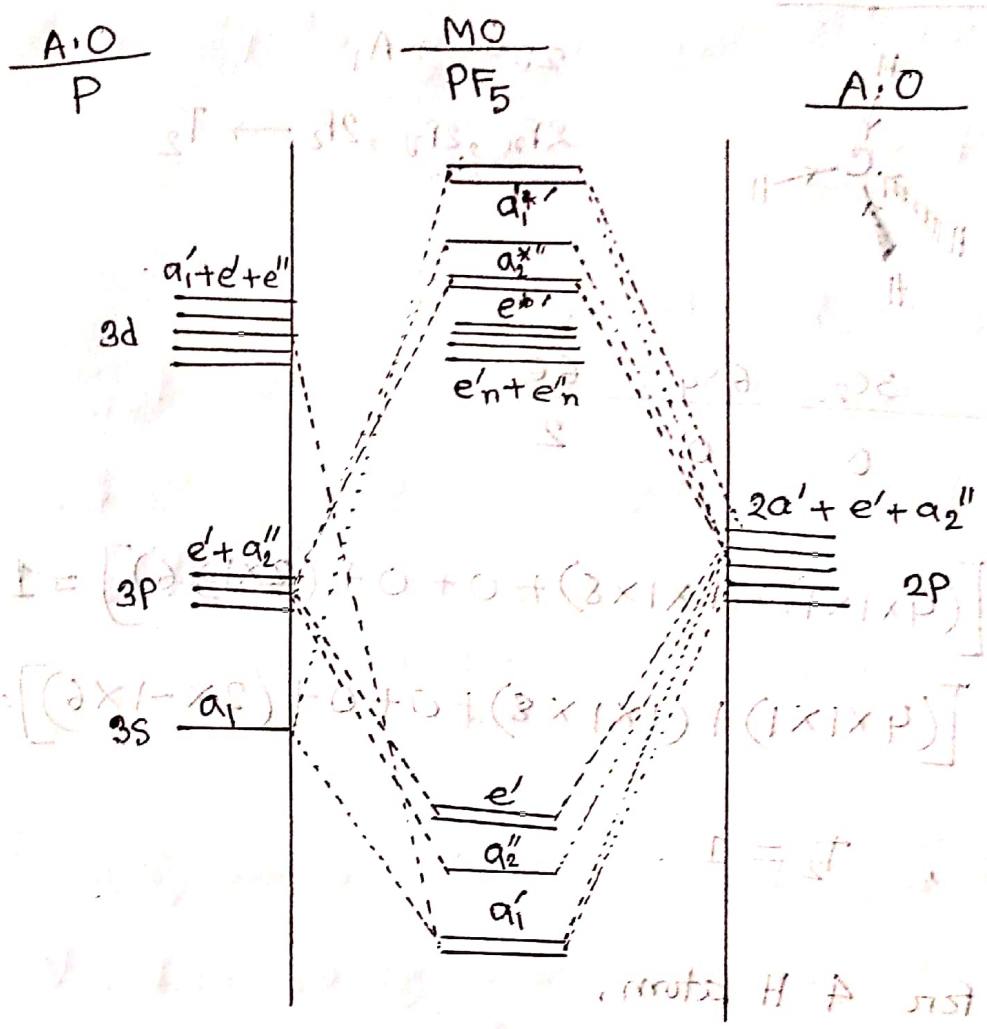
| $D_{3h}$ | E | $2C_3$ | $3C_2$ | $\sigma_h$ | $2S_3$ | $3\sigma_v$ |
|---|---|---|---|---|---|---|
| $\Gamma$ | 5 | 2 | 1 | 3 | 0 | 3 |

2 marks

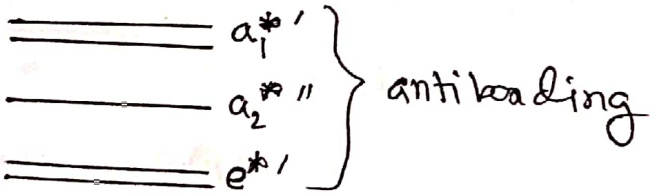
No. of $A_1' = \frac{1}{12}[(5\times1\times1)+(2\times1\times2)+(1\times1\times3)+(3\times1\times1)+0+(3\times1\times3)]$

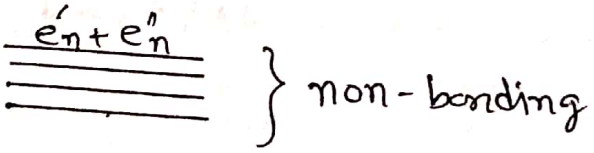
$= 2$

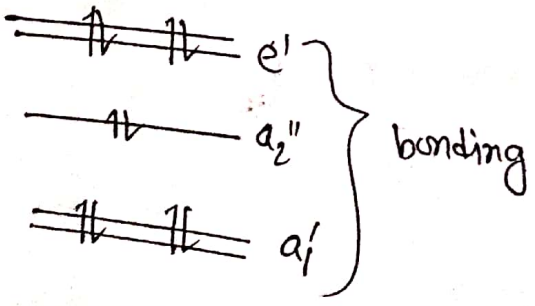
$\Gamma = 2A_1' + E' + A_2''$ ] For '5' F




P has 5 valence electron & each F has one valen electron. Total (5+5) = 10.



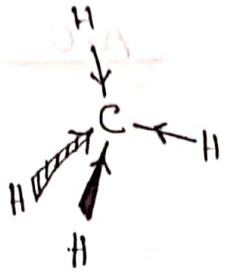




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## $CH_4$ Tetrahedral $T_d$ :

Structure :




$2S \longrightarrow A_1$

$2P_x, 2P_y, 2P_z \longrightarrow T_2$

| $T_d$ | E | $8C_3$ | $3C_2$ | $6S_4$ | $6\sigma_d$ |
|---|---|---|---|---|---|
| $\Gamma$ | 4 | 1 | 0 | 0 | 2 |

No. of $A_1 = \frac{1}{24}\left[(4\times1\times1)+(1\times1\times8)+0+0+(2\times1\times6)\right] = 1$

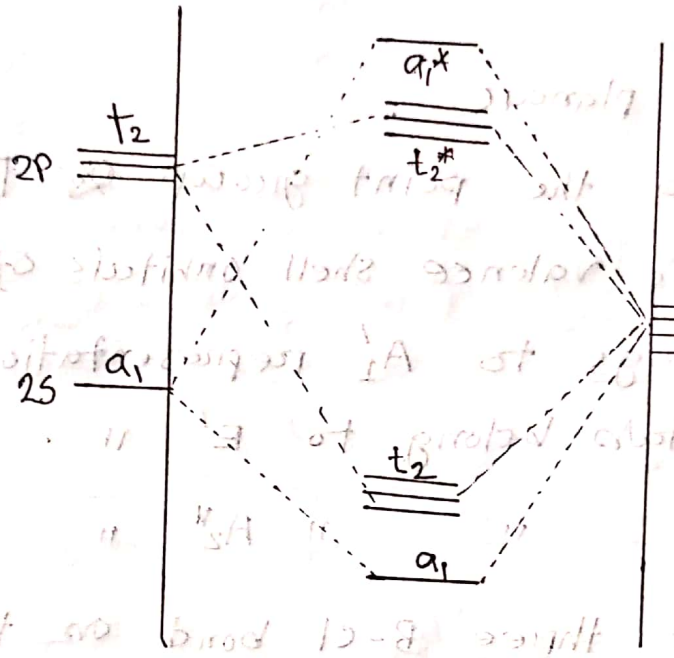
No. of $A_2 = \frac{1}{24}\left[(4\times1\times1)+(1\times1\times8)+0+0+(2\times-1\times6)\right] = 0$

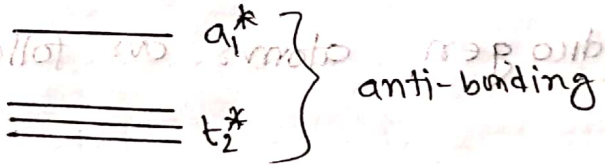
$E = 0, \quad T_1 = 0, \quad T_2 = 1.$

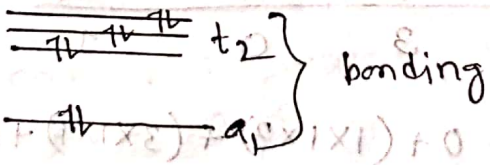
$\Gamma = A_1 + T_2$ ] for 4 H atom.

A.O | M.O. | A.O

C | $CH_4$ | H



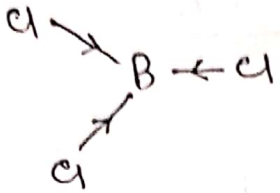

Bond order = 4.

∴ So, there are 4, C-H bond. No more bonding electron.

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## $BCl_3$ :

The structure of $NH_3$ is trigonal planar.




Trigonal planar :

$BCl_3$ belongs to the point group $D_{3h}$

Among the three valence shell orbitals of B.

2s orbital belongs to $A_1'$ representation.

$2P_x$ and $2P_y$ orbitals belong to $E'$ ".

$2P_z$ " " " $A_2''$ "

By taking the three B-Cl bond as the basis, we can calculate the group orbitals for the 3-hydrogen atoms as follows.

| $D_{3h}$ | E | $2C_3$ | $3C_2$ | $\sigma_h$ | $2S_3$ | $3\sigma_v$ |
|---|---|---|---|---|---|---|
| $\Gamma$ | 3 | 0 | 1 | 3 | 0 | 1 |

No. of $A_1' = \frac{1}{12}[(3\times1\times1)+0+(1\times1\times3)+(3\times1\times1)+0+(1\times1\times3)] = 1$

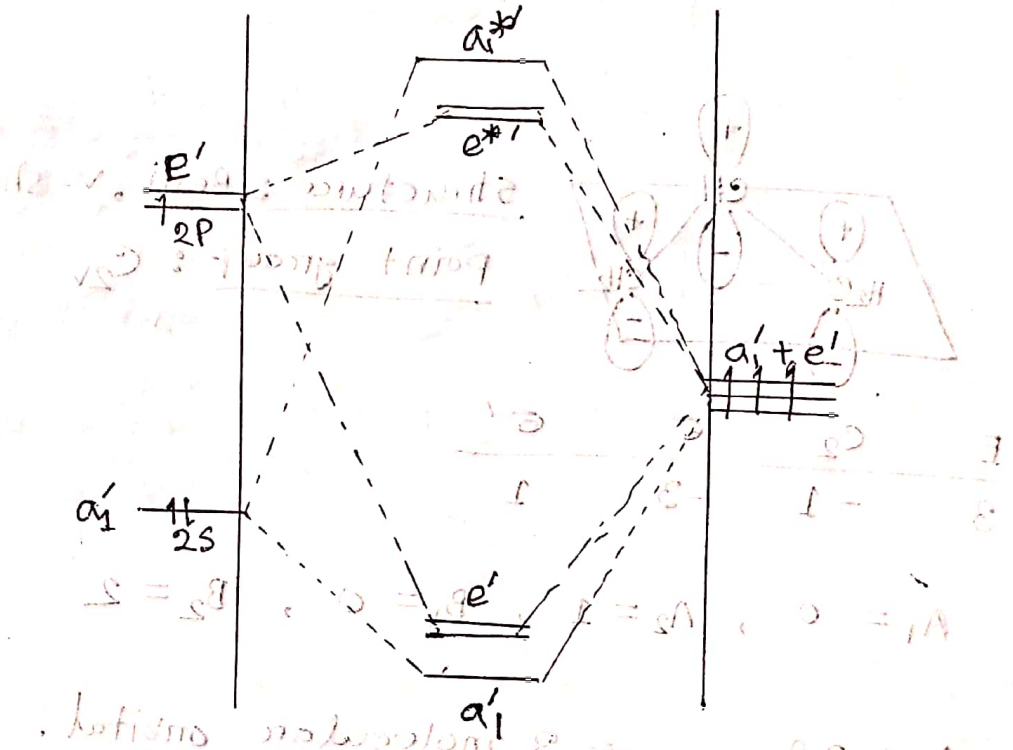
" " $A_2' = \frac{1}{12}[(3\times1\times1)+0+(1\times-1\times3)+(3\times1\times1)+0+(1\times-1\times3)] = 0$

" " $E' = \frac{1}{12}[(3\times2\times1)+0+(1\times0\times3)+(3\times2\times1)+0+0] = 1$

No of $A_1'' = 0$, $A_2'' = 0$, $E'' = 0$

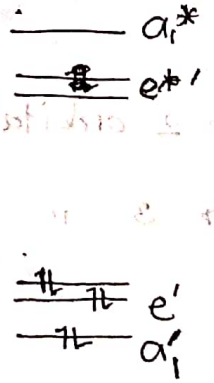
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$\Gamma = A_1' + E'$




In, $BCl_3$, B has 3 electrons in its valence shell and 3 hydrogen atom have 3 electrons.

Now we can put theseese $3+3=6$ electrons into the molecular orbitals as follows.




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$CH_2 \cdots\cdots CH \cdots\cdots CH_2$

$H_2C = CH - \overset{\oplus}{C}H_2$

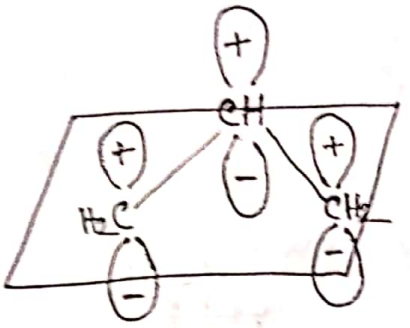
$H_2C = CH - CH_2^{\bullet}$

$CH_2 = CH - CH_2 - Cl$

Alkyl chloride.

Testing

3




Structure : Bent, V-shape

point group : $C_{2v}$

| $C_{2v}$ | E | $C_2$ | $\sigma$ | $\sigma'$ |
|---|---|---|---|---|
| $\Gamma$ | 3 | -1 | -3 | 1 |

No. of $A_1 = 0$, $A_2 = 1$, $B_1 = 0$, $B_2 = 2$

So, $\Gamma = A_2 + 2B_2$ ⟶ 3 molecular orbital.

Two different energy state $A_2$ & 2 $B_2$

⟶ $2B_2$ means 2 orbitals of same symmetry but energy different.

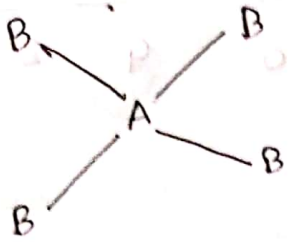
যদি energy same হতো তাহলে সেটা E হতো, $2B_2$ লিখতে হতো না।

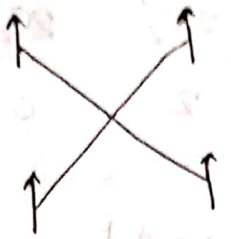
⟶ E → double degenerate → 2 orbitals of same energy.

⟶ T → Triple degenerate ⟶ 3 " " " "

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Test: 1 & 2



[Arrows are out of plane]

| $D_{4h}$ | E | $2C_4$ | $C_2$ | $2C_2'$ | $2C_2''$ | i | $2S_4$ | $\sigma_h$ | $2\sigma_v$ | $2\sigma_d$ |
|---|---|---|---|---|---|---|---|---|---|---|
| $\Gamma$ | 4 | 0 | 0 | -2 | 0 | 0 | 0 | -4 | 2 | 0 |

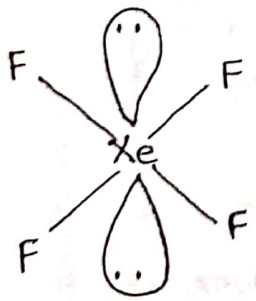
$\Gamma = E_g + A_{2u} + B_{2u}$ [suitable for $\pi$-bonding]

Suitables orbital for out of plane $\pi$-bonding are → $E_g \rightarrow d_{xz}, d_{yz}$; $A_{2u} \rightarrow p_z$; $B_{2u} \rightarrow$ none.

কোন atom center atom (main or transition), তার উপর base করে orbital গুলো overlap করবে। main (p-orbital), transition (d-orbital).

$XeF_4$: $^{54}Xe \rightarrow {}^{36}[Kr]\ 4d^{10}\ 5s^2\ 5p^6\ 5d^0$

$= {}^{36}[Kr]\ \underbrace{5s^2\ 5p_x^2\ 5p_y^1\ 5p_z^1\ 5d^1\ 5d^1\ 5d^0\ 5d^0\ 5d^0}_{sp^3d^2}$




$5d_{xz}$ অথবা $5d_{yz}$ এর যেকোন একটি ব্যবহার করে $\pi$-bond form করতে পারবে।

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| $D_{4h}$ | E | $2C_4$ | $C_2$ | $2C_2'$ | $2C_2''$ | i | $2S_4$ | $\sigma_h$ | $2\sigma_v$ | $2\sigma_d$ |
|---|---|---|---|---|---|---|---|---|---|---|
| $\Gamma$ | 4 | 0 | 0 | 2 | 0 | 0 | 0 | 4 | 2 | 0 |

In plane : $\Gamma = A_{1g} + B_{1g} + E_u$

$A_{1g} \longrightarrow s, d_{z^2}$

$B_{1g} \longrightarrow d_{x^2-y^2}$

$E_u \longrightarrow p_x, p_y$

$dsp^2$ / $sp^2d$

$A_{1g}$ is single degenerate and ~~by~~ highly degenerate, bez all operations are same & it is 1. so, it represent either 's' or $d_{z^2}$.



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# Programme : 07

## * Application of Molecular Vibration *

7.2 : Group theory can be an enormous help in deciding the infrared or Raman activity of different molecular vibrations.

→ But before considering spectra we must look more generally at the subject of vibrations.

Any movement of an atom in a molecule can be resolved into three components along the x, y and z axes.

→ If therefore, there are 'n' atoms in a molecule, there are '3n' ~~atoms in a mol.~~ possible movements of its atoms.

- 3 will be concerted movements of the whole molecule along the three co-ordinate axes, i.e.. "translations".
- 3 (or 2 for a linear molecule) will be concerned "rotations" about the axes.
- The remaining 3n-6 (or 3n-5 for a linear molecule) must therefore be "molecular vibrations."

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- How many vibrations will there ~~for~~ be for the molecule? $XeF_4$?

→ There are 5 atoms, $n = 5$

So, total movement of the atoms of the molecule is $3n = 3 \times 5 = 15$.

→ This is not a linear molecule. So, 3 translations, '3' rotation,

So, the remaining $= 3n - 6$

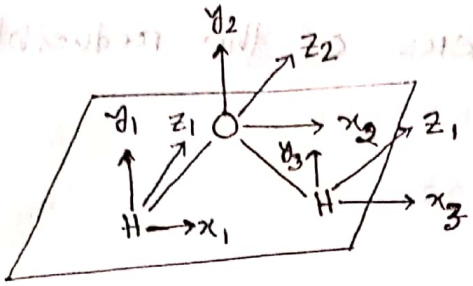
$= 3 \times 5 - 6$

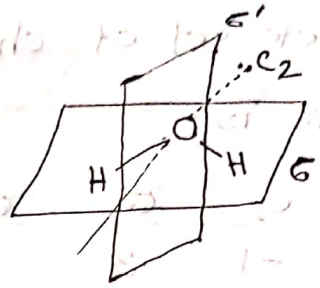
$= 9$ ; are for vibrations.

7.3 We can find the symmetries of all the possible molecular motions by using x, y & z directions on each atom has a basis for a reducible representation of the group.

- For an n-atom molecule, this will produce a representation ~~will be 3n~~ of order 3n, i.e., the character of the identity representation will be 3n. all the matriexs involved will be $3n \times 3n$ matrices.
- The quick way to finding the character of a matrix is equal to the extent to which the vectors in the basis are left unshifted by the operation.

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Group : $C_{2v}$ Symmetry Operations : $C_2, E, \sigma, \sigma'$

Characters of each operations,

→ E: $\chi = 9$ (all arrows unshifted).

→ $C_2$ : $\chi = -1$ (all arrows on atoms 1 & 3 are shifted).

Then, Due to $C_2$ operation,

$x_2$ becomes $-x_2$

$y_2$ " $-y_2$

$z_2$ " $+z_2$

overall unshifted $= -1$ $(-2+1=-1)$.

→ $\sigma$ : $\chi = 3$

all x and z are unshifed, so, character = 6

all y becomes $-y$, so character $= -3$.

Overall $= (6-3) = 3$.

→ $\sigma'$ : $\chi = 1$ ; (1 & 3 are shifted).

$x_2$ becomes $-x_2$, so, $-1$

$y_2, z_2$ are unshifted, so $+2$

overall $= (+2-1) = +1$.

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Thus the complete set of characters of the reducible representation is:

| $C_{2v}$ | E | $C_2$ | $\sigma$ | $\sigma'$ |
|---|---|---|---|---|
| $\Gamma$ | 9 | -1 | 3 | 1 |

Reducible representation $\Gamma_1 = 3A_1 + A_2 + 3B_1 + 2B_2$

| $C_{2v}$ | E | $C_2$ | $\sigma_v(xy)$ | $\sigma'_v(yz)$ | | |
|---|---|---|---|---|---|---|
| $A_1$ | 1 | 1 | 1 | 1 | z | $x^2, y^2, z^2$ |
| $A_2$ | 1 | 1 | -1 | -1 | $R_z$ | xy |
| $B_1$ | 1 | -1 | 1 | -1 | x, $R_y$ | xz |
| $B_2$ | 1 | -1 | -1 | 1 | y, $R_x$ | yz |

$F_1 = 3A_1 + A_2 + 3B_1 + 2B_2$

→ These are the symmetry species of all nine possible molecular movements.

→ From these nine we must now remove the translations and rotations.

→ The **translations** must belong to $A_1$, $B_1$ & $B_2$ because they must be affected by the group operations in the same way as the x, y & z directions.

→ **Rotations** belongs to $A_2$, $B_1$, $B_2$ ($R_z$, $R_y$, $R_x$ in the character table).

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- Remove translations & Rotations from the nine species we get,

$2A_1 + B_1$

→ These are the symmetries of the three vibrational modes of the water molecule (or of any other triatomic $C_{2v}$ molecule).

- Summarise

Total = $3A_1 + A_2 + 3B_1 + 2B_2$

Translations = $A_1 + B_1 + B_2$

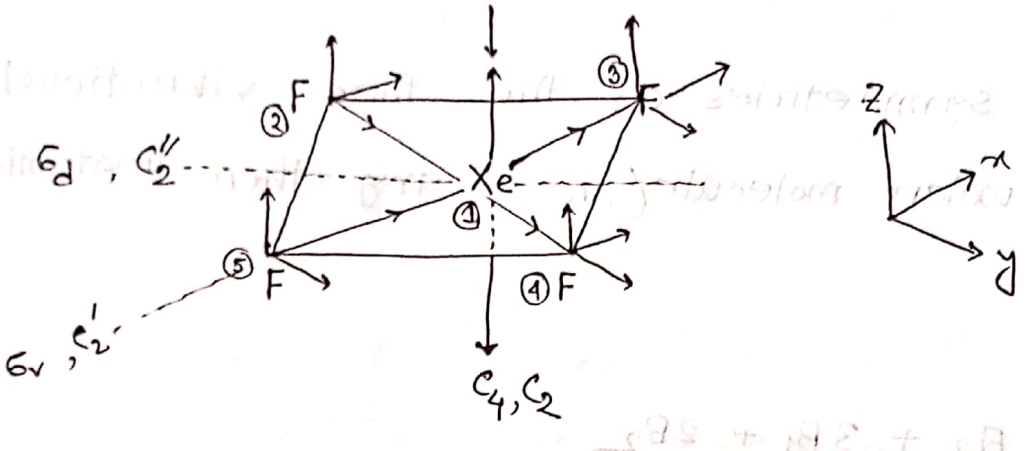
Rotations = $A_2 + B_1 + B_2$

Vibrations = $2A_1 + B_1$

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## $XeF_4$ $D_{4h}$ Group, Square planar.

5 atoms, $n=5$, so, $3\times5 = 15$ mov




| $D_{4h}$ | E | $2C_4$ | $C_2(C_4^2)$ | $2C_2'$ | $2C_2''$ | i | $2S_4$ | $\sigma_h$ | $2\sigma_v$ | $2\sigma_d$ |
|---|---|---|---|---|---|---|---|---|---|---|
| $\Gamma$ | 15 | 1 | −1 | −3 | −1 | −3 | −1 | 5 | 3 | 1 |

**E = 15**

**$C_4$ ; (2,3,4,5) shifted**

$z = +z$

$x = +y$

$y = -x$

Net = +1

**$C_2$ ; $z = +z$**

$x = -x$

$y = -y$

Net = −1

**$C_2'$ ; (2,4) shifted**

5,1,3; 3 x unshifted

others ; 3z, 3y shifted

Net = (−6+3) = −3

**$C_2''$ ; (2,3,4,5) Shifted.**

$x = +x$

$y = -y$

$z = -z$

Net = −1

**i ; $x = -x$**

$y = -y$

$z = -z$

Net = −3

**$S_4 = -1$**

**$\sigma_h$ ; x,y in plane**

z, out of "

10, x, y ~~wash~~ = +

−5 ; z change direct

Net = +5

**$\sigma_v$ ; the plane contain x, z.**

6 x, z don't change direction

(−) 3 y " "

Net = +3

**$\sigma_d$ ; x, z = +**

y = −

Net = +1

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## Reducible Representation:

$D_{4h}$

$\Gamma = A_{1g} + A_{2g} + B_{1g} + B_{2g} + E_g + 2A_{2u} + B_{2u} + 3E_u$

Total degeneracy of '$\Gamma$' is **15**.

→ The degeneracy equals the number of vectors in the original basis. This is always true.

→ 15 degeneracy
- → 3 translation → 3 rotations.

So, vibrational modes → $(3n-6) = (15-6) = 9$.

→ Symmetry species of the translations: $A_{2u} + E_u$
- → $(A_{2u})$: singly degenerate translation along 'z'
- → $(E_u)$: two equivalent translations along x & y

→ Symmetry species of the rotations: $A_{2g} + E_g$
- $A_{2g}$ → $R_z$
- $E_g$ → $(R_x, R_y)$

$\Gamma = A_{1g} + A_{2g} + B_{1g} + B_{2g} + E_g + 2A_{2u} + B_{2u} + 3E_u$

$A_{2u} + E_u$

Translations = $A_{2u} + E_u$

Rotations = $A_{2g} + E_g$

Vibrations = $A_{1g} + B_{1g} + B_{2g} + A_{2u} + B_{2u} + 2E_u$

Total degeneracy = 9 for 9 vibration $(3n-6)$.

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"Breathing" mode in which all four fluorines move out together and then in together. This mode of vibration clearly maintains the full symmetry of the molecule and therefore belongs to the $A_{1g}$ irreducible representation.

## IR & Raman active spectra:

We now need to determine which, if any, of these modes of vibrations are active in the Infrared & Raman spectra of the molecule.

## Rules:

① A vibration will be infrared active if it belongs to the same symmetry species as a component of dipole moment, i.e. to the same species as either x, y, or z.

For $XeF_4$ : $A_{2u} + 2E_u$ are IR active.

for $H_2O$ : $2A_1 + B_1$ " " " ,

→ In both molecules there should be three IR active bands.

→ $2A_1$ indicate two different vibrations (no degenerate) of the same symmetry.

→ $2E_u$ indicates again two bonds, but each one consists of two degenerate vibrations.

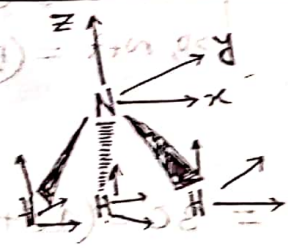
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## Raman Rule:

(ii) A fundamental vibration will be Raman active, if it belongs to the same symmetry species as a component of polarisability, i.e. to one of the binary products, $x^2, y^2, z^2, xy, xz, yz$ or to a combination of products such as $x^2 - y^2$.

- $H_2O$ : all three are active bcz $x^2, y^2$ & $z^2$ belongs to $A_1$ & $xz$ belongs to $B_1$.
- $XeF_4$ : $A_{1g}, B_{1g}, B_{2g}$ are Raman active.

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- 


$C_{3v}$ point group

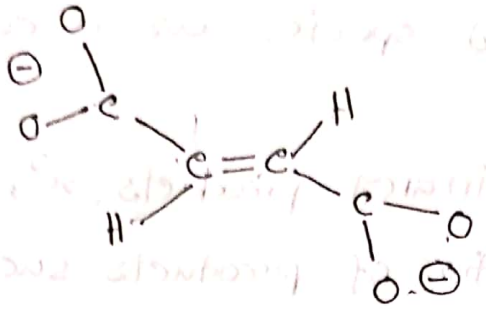
| $C_{3v}$ | E | $2C_3$ | $3\sigma_v$ |
|---|---|---|---|
| $\Gamma$ | 12 | 0 | 2 |

$\sigma_v$ : ( x & z are unshifted on two atoms, y becomes -y).

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(1)



$C_{2h}$ group.

n = 10 atom.

Take 3×10 arrows as basis.

Reducible Representation.

| $C_{2h}$ | E | $C_2$ | i | $\sigma_h$ |
|---|---|---|---|---|
| Γ | 30 | 0 | 0 | 10 |

$\sigma_h$

10 x, 10 y lay in plane, 10 z perpendicular

20 arrow doesn't change direction but 10 arrows change.

So, Net = (20 − 10) = 10

No. of Ag = $\frac{1}{4}[(30\times1\times1)+0+0+(10\times1\times1)]$

= 10

$\Gamma = 10A_g + 5B_g + 5A_u + 10B_u$.

Total mode of vibration = 3n = 30 = (10+5+5+10).

Translations No = $A_u + 2B_u$

Rotations = $A_g + 2B_g$.

So, Vibrations = $9A_g + 3B_g + 4A_u + 8B_u$.

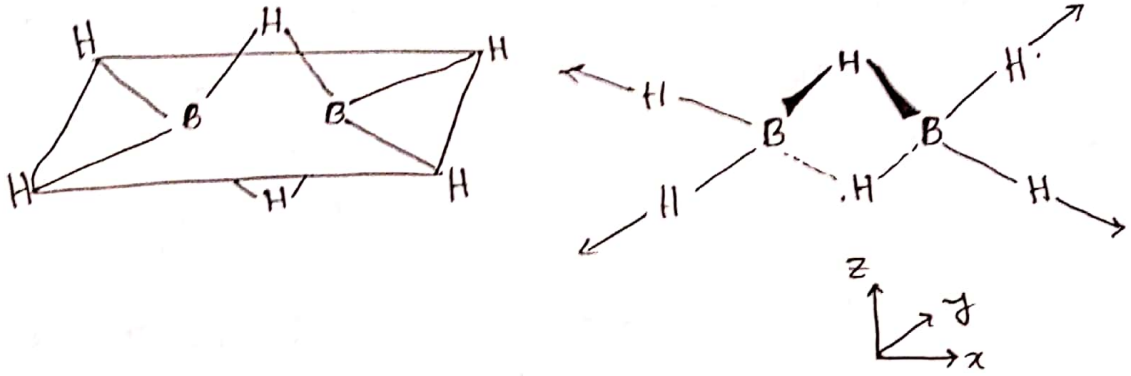
IR active = $4A_u + 8B_u$

Raman active = $9A_g + 3B_g$.

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(2) OK

(3) Terminal B–H stretching vibrations.




| $D_{2h}$ | E | $C_2(z)$ | $C_2(y)$ | $C_2(x)$ | i | $\sigma(xy)$ | $\sigma(xz)$ | $\sigma(yz)$ |
|---|---|---|---|---|---|---|---|---|
| $\Gamma$ | 4 | 0 | 0 | 0 | 0 | 4 | 0 | 0 |

irreducible representation $\Gamma = A_g + B_{1g} + B_{2u} + B_{3u}$

IR active ; $B_{2u} + B_{3u}$.

Raman active, $A_g + B_{1g}$.

} Different spectra কারণ symmetry different.

The End